



Electrostatics : RACE-01

1. A $10 \mu\text{C}$ charge is divided into two parts and placed at 1 cm distance so that the repulsive force between them is maximum. The charges of the two parts are :

(A) $9 \mu\text{C}, 1 \mu\text{C}$ (B) $5 \mu\text{C}, 5 \mu\text{C}$ (C) $7 \mu\text{C}, 3 \mu\text{C}$ (D) $8 \mu\text{C}, 2 \mu\text{C}$

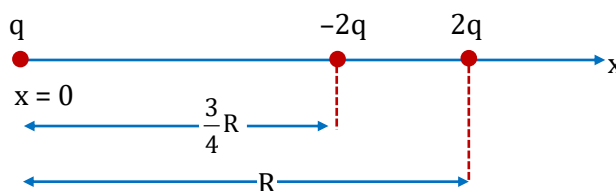
PEO313

2. Three charges $+Q, q, +Q$ are placed respectively, at distance, $0, d/2$ and d from the origin, on the x -axis. If the net force experienced by $+Q$, placed at $x = 0$, is zero, then value of q is :

(A) $+Q/2$ (B) $-Q/2$ (C) $-Q/4$ (D) $+Q/4$

PEO314

3. Three point charges $q, -2q$ and $2q$ are placed on x -axis at a distance $x = 0, x = \frac{3}{4}R$ and $x = R$ respectively from origin as shown. If $q = 2 \times 10^{-6} \text{ C}$ and $R = 2 \text{ cm}$, the magnitude of net force experienced by the charge $-2q$ is _____ N .



PEO315

4. An infinite number of point charges, each carrying $1 \mu\text{C}$ charge, are placed along the y -axis at $y = 1 \text{ m}, 2 \text{ m}, 4 \text{ m}, 8 \text{ m}$ _____. The total force on a 1 C point charge, placed at the origin, is $x \times 10^3 \text{ N}$. The value of x , to the nearest integer, is _____.

[Take $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2/\text{C}^2$]

PEO316

5. Force between two point charges q_1 and q_2 placed in vacuum at ' r ' cm apart is F . Force between them when placed in a medium having dielectric $K = 5$ at ' $r/5$ ' cm apart will be:

(A) $F/25$ (B) $5F$ (C) $F/5$ (D) $25F$

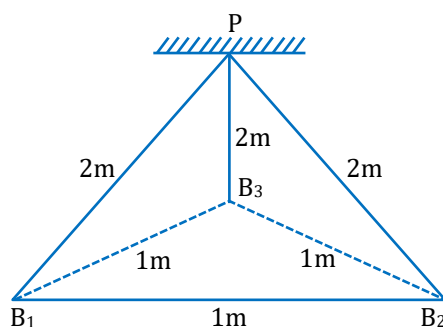
PEO317

6. If two charges q_1 and q_2 are separated with distance ' d ' and placed in a medium of dielectric constant K . What will be the equivalent distance between charges in air for the same electrostatic force ?

(A) $d\sqrt{k}$ (B) $k\sqrt{d}$ (C) $1.5d\sqrt{k}$ (D) $2d\sqrt{k}$

PEO318

7. Three identical charged balls each of charge $2C$ are suspended from a common point P by silk threads of $2m$ each (as shown in figure). They form an equilateral triangle of side $1m$. The ratio of net force on a charged ball to the force between any two charged balls will be :



- (A) $1 : 1$ (B) $1 : 4$ (C) $\sqrt{3} : 2$ (D) $\sqrt{3} : 1$

PEO319

8. Two identical charged spheres are suspended by strings of equal lengths. The strings make an angle θ with each other. When suspended in water the angle remains the same. If density of the material of the sphere is 1.5 g/cc , the dielectric constant of water will be _____.
(Take density of water = 1 g/cc)

PEO320

9. Two equal positive point charges are separated by a distance $2a$. The distance of a point from the centre of the line joining two charges on the equatorial line (perpendicular bisector) at which force experienced by a test charge q_0 becomes maximum is $\frac{a}{\sqrt{x}}$. The value of x is _____.

PEO321

10. A thin metallic wire having cross sectional area of 10^{-4} m^2 is used to make a ring of radius 30 cm . A positive charge of $2\pi \text{ C}$ is uniformly distributed over the ring, while another positive charge of 30 pC is kept at the centre of the ring. The tension in the ring is _____ N ; provided that the ring does not get deformed (neglect the influence of gravity). (given, $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ SI units}$)

PEO322

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	C	5440	12	B	A	D	3	2	3



Electrostatics : RACE-02

1. Two charges q and $3q$ are separated by a distance ' r ' in air. At a distance x from charge q , the resultant electric field is zero. The value of x is :

(A) $\frac{(1+\sqrt{3})}{r}$ (B) $\frac{r}{3(1+\sqrt{3})}$ (C) $\frac{r}{(1+\sqrt{3})}$ (D) $r(1+\sqrt{3})$

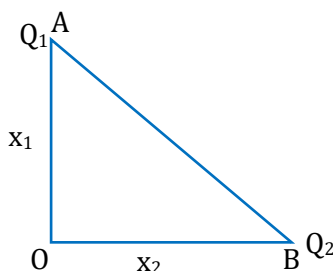
PEO323

2. Two point charges $q_1(\sqrt{10} \mu C)$ and $q_2(-25 \mu C)$ are placed on the x -axis at $x = 1 \text{ m}$ and $x = 4 \text{ m}$ respectively. The electric field (in V/m) at a point $y = 3 \text{ m}$ on y -axis is, $\left[\text{take } \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2\text{C}^{-2} \right]$

(A) $(-63\hat{i} + 27\hat{j}) \times 10^2$ (B) $(81\hat{i} - 81\hat{j}) \times 10^2$
 (C) $(63\hat{i} - 27\hat{j}) \times 10^2$ (D) $(-81\hat{i} + 81\hat{j}) \times 10^2$

PEO324

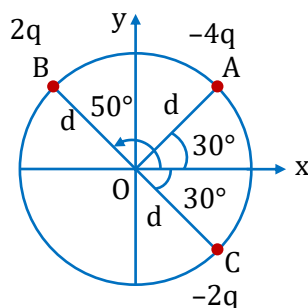
3. Charges Q_1 and Q_2 are at points A and B of a right angle triangle OAB (see figure). The resultant electric field at point O is perpendicular to the hypotenuse, then Q_1/Q_2 is proportional to :



(A) $\frac{x_2^2}{x_1^2}$ (B) $\frac{x_1^3}{x_2^3}$ (C) $\frac{x_1}{x_2}$ (D) $\frac{x_2}{x_1}$

PEO325

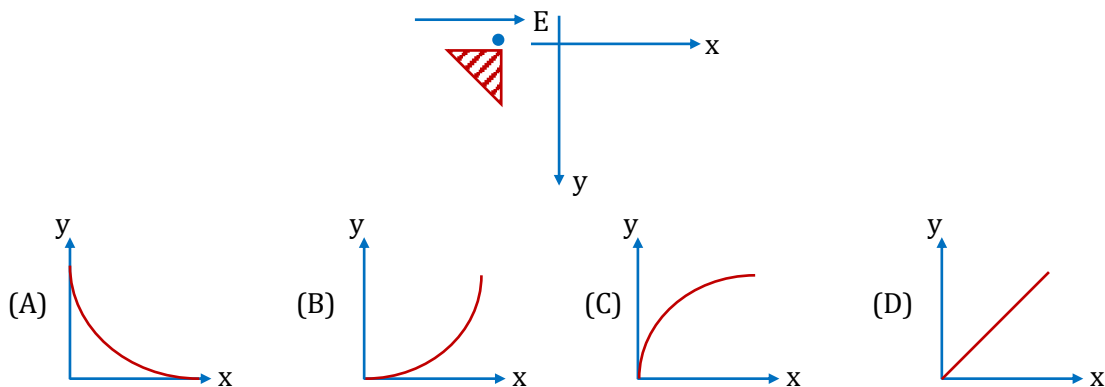
4. Three charged particle A, B and C with charges $-4q, 2q$ and $-2q$ are present on the circumference of a circle of radius d . the charged particles A, C and centre O of the circle formed an equilateral triangle as shown in figure. Electric field at O along x -direction is :



(A) $\frac{2\sqrt{3}q}{\pi\epsilon_0 d^2}$ (B) $\frac{\sqrt{3}q}{4\pi\epsilon_0 d^2}$ (C) $\frac{3\sqrt{3}q}{4\pi\epsilon_0 d^2}$ (D) $\frac{\sqrt{3}q}{\pi\epsilon_0 d^2}$

PEO326

5. A small point mass carrying some positive charge on it, is released from the edge of a table. There is a uniform electric field in this region in the horizontal direction. Which of the following options then correctly describe the trajectory of the mass ? (Curves are drawn schematically and are not to scale).

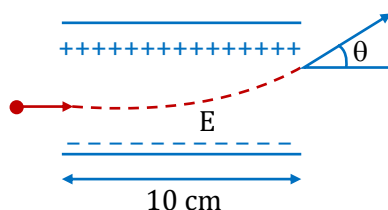


PEO327

6. A stream of a positively charged particles having $\frac{q}{m} = 2 \times 10^{11} \frac{C}{kg}$ and velocity $\vec{v}_0 = 3 \times 10^7 \hat{i} \text{ m/s}$ is deflected by an electric field $1.8 \hat{j} \text{ V/m}$. The electric field exists in a region of 10 cm along x direction. Due to the electric field, the deflection of the charge particles in the y -direction is _____ mm .

PEO328

7. A uniform electric field of 10 N/C is created between two parallel charged plates (as shown in figure). An electron enters the field symmetrically between the plates with a kinetic energy 0.5 eV . The length of each plate is 10 cm . The angle (θ) of deviation of the path of electron as it comes out of the field is _____ (in degree).

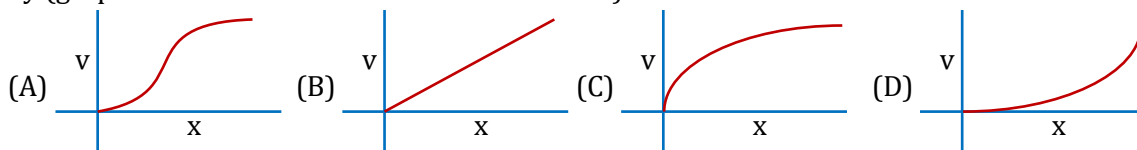


PEO329

8. The bob of a simple pendulum has mass $2g$ and a charge of $5.0 \mu\text{C}$. It is at rest in a uniform horizontal electric field of intensity 2000 V/m . At equilibrium, the angle that the pendulum makes with the vertical is: (take $g = 10 \text{ m/s}^2$)
 (A) $\tan^{-1}(5.0)$ (B) $\tan^{-1}(2.0)$ (C) $\tan^{-1}(0.5)$ (D) $\tan^{-1}(0.2)$

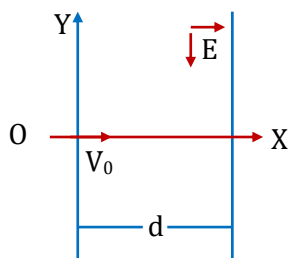
PEO330

9. A particle of mass m and charge q is released from rest in a uniform electric field. If there is no other force on the particle, the dependence of its speed v on the distance x travelled by it is correctly given by (graphs are schematic and not drawn to scale)



PEO331

10. A charged particle (mass m and charge q) moves along X axis with velocity V_0 . When it passes through the origin it enters a region having uniform electric field $\vec{E} = -E\hat{j}$ which extends upto $x = d$. Equation of path of electron in the region $x > d$ is :



- (A) $y = \frac{qEd}{mV_0^2} \left(\frac{d}{2} - x \right)$ (B) $y = \frac{qEd}{mV_0^2} (x - d)$
- (C) $y = \frac{qEd}{mV_0^2} x$ (D) $y = \frac{qEd^2}{mV_0^2} x$

PE0332

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	C	C	C	D	D	2	45	C	C	A



Electrostatics : RACE-03

1. For a uniformly charged ring of radius R , the electric field on its axis has the largest magnitude at a distance h from its centre. Then value of h is :

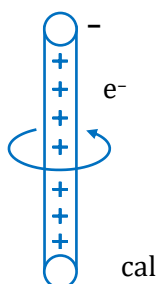
(A) $\frac{R}{\sqrt{5}}$ (B) R (C) $\frac{R}{\sqrt{2}}$ (D) $R\sqrt{2}$

PE0333

2. Suppose a uniformly charged wall provides a uniform electric field of $2 \times 10^4 \text{ N/C}$ normally. A charged particle of mass 2 g being suspended through a silk thread of length 20 cm and remain stayed at a distance of 10 cm from the wall. Then the charge on the particle will be $\frac{1}{\sqrt{x}} \mu\text{C}$ where $x = \underline{\hspace{1cm}}$. [use $g = 10 \text{ m/s}^2$]

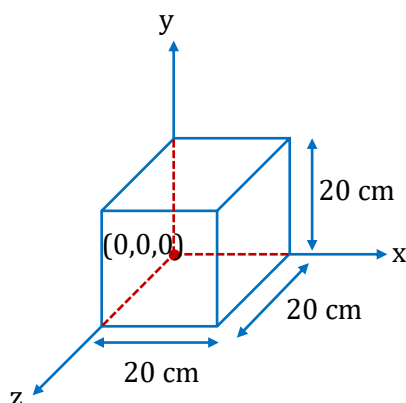
PE0334

3. An electron revolves around an infinite cylindrical wire having uniform linear charge density $2 \times 10^{-8} \text{ Cm}^{-1}$ in circular path under the influence of attractive electrostatic field as shown in the figure. The velocity of electron with which it is revolving is $\underline{\hspace{1cm}} \times 10^6 \text{ ms}^{-1}$. Given mass of electron $= 9 \times 10^{-31} \text{ kg}$.



PE0335

4. Expression for an electric field is given by $\vec{E} = 4000x^2 \hat{i} \frac{\text{V}}{\text{m}}$. The electric flux through the cube of side 20 cm when placed in electric field (as shown in the figure) is $\underline{\hspace{1cm}} \text{ V cm}$.



PE0336

5. An electric field is given by $(6\hat{i} + 5\hat{j} + 3\hat{k}) \text{ N/C}$. The electric flux through a surface area $30\hat{i} \text{ m}^2$ lying in YZ -plane (in SI unit) is :

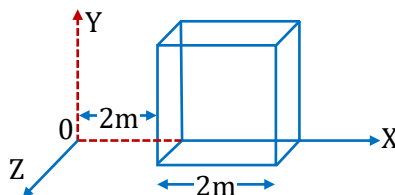
(A) 90 (B) 150 (C) 180 (D) 60

PEO337

6. An electric field, $\vec{E} = \frac{2\hat{i} + 6\hat{j} + 8\hat{k}}{\sqrt{6}}$ passes through the surface of 4 m^2 area having unit vector $\hat{n} = \left(\frac{2\hat{i} + \hat{j} + \hat{k}}{\sqrt{6}}\right)$. The electric flux for that surface is _____ Vm .

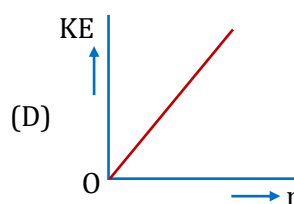
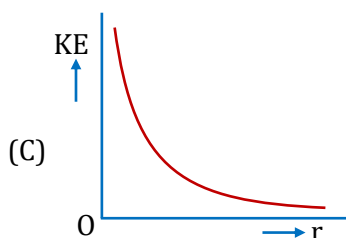
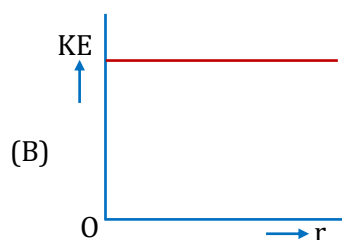
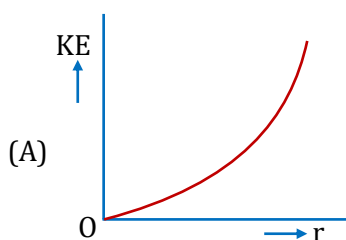
PEO338

7. An electric field $\vec{E} = (2x\hat{i}) \text{ NC}^{-1}$ exists in space. A cube of side 2m is placed in the space as per figure given below. The electric flux through the cube is Nm^2/C .



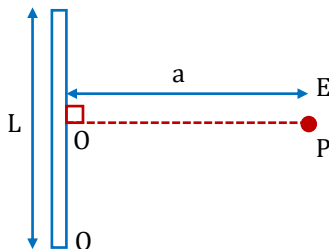
PEO339

8. An infinitely long positively charged straight thread has a linear charge density $\lambda \text{ Cm}^{-1}$. An electron revolves along a circular path having axis along the length of the wire. The graph that correctly represents the variation of the kinetic energy of electron as a function of radius of circular path from the wire is :



PEO340

9. Find the electric field at point P (as shown in figure) on the perpendicular bisector of a uniformly charged thin wire of length L carrying a charge Q . The distance of the point P from the centre of the rod is $a = \frac{\sqrt{3}}{2}L$.



- (A) $\frac{\sqrt{3}Q}{4\pi\epsilon_0 L^2}$ (B) $\frac{Q}{3\pi\epsilon_0 L^2}$ (C) $\frac{Q}{2\sqrt{3}\pi\epsilon_0 L^2}$ (D) $\frac{Q}{4\pi\epsilon_0 L^2}$

PEO341

10. A particle of charge $-q$ and mass m moves in a circle of radius r around an infinitely long line charge of linear density $+\lambda$. Then time period will be given as:
(Consider k as Coulomb's constant)

- (A) $T^2 = \frac{4\pi^2 m}{2k\lambda q} r^3$ (B) $T = 2\pi r \sqrt{\frac{m}{2k\lambda q}}$ (C) $T = \frac{1}{2\pi r} \sqrt{\frac{m}{2k\lambda q}}$ (D) $T = \frac{1}{2\pi} \sqrt{\frac{2k\lambda q}{m}}$

PEO342

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	C	3	8	640	C	12	16	B	C	B



Electrostatics : RACE-04

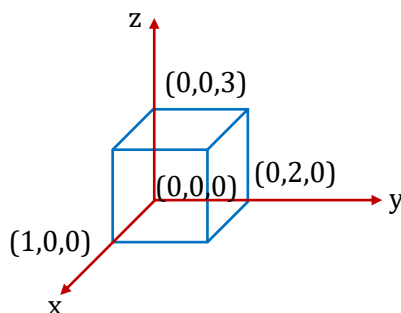
1. In a cuboid of dimension $2L \times 2L \times L$, a charge q is placed at the centre of the surface 'S' having area of $4L^2$. The flux through the opposite surface to 'S' is given by

(A) $\frac{q}{12\epsilon_0}$ (B) $\frac{q}{3\epsilon_0}$ (C) $\frac{q}{2\epsilon_0}$ (D) $\frac{q}{6\epsilon_0}$

PEO343

2. As shown in figure, a cuboid lies in a region with electric field $E = 2x^2\hat{i} - 4y\hat{j} + 6\hat{k}$ N/C. The magnitude of charge within the cuboid is $n\epsilon_0 C$. The value of n is _____.

(if dimension of cuboid is $1 \times 2 \times 3 m^3$)



PEO344

3. A cubical volume is bounded by the surfaces $x = 0, x = a, y = 0, y = a, z = 0, z = a$. The electric field in the region is given by $\vec{E} = E_0 x \hat{i}$. Where $E_0 = 4 \times 10^4 NC^{-1} m^{-1}$. If $a = 2 cm$, the charge contained in the cubical volume is $Q \times 10^{-14} C$. The value of Q is _____. (Take: $\epsilon_0 = 9 \times 10^{-12} C^2/Nm^2$)

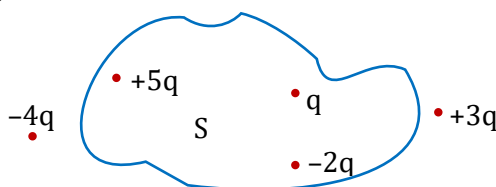
PEO345

4. A charge q is placed at the center of one of the surface of a cube. The flux linked with the cube is :-

(A) $\frac{q}{4\epsilon_0}$ (B) $\frac{q}{2\epsilon_0}$ (C) $\frac{q}{8\epsilon_0}$ (D) Zero

PEO346

5. Five charges $+q, +5q, -2q, +3q$ and $-4q$ are situated as shown in the figure. The electric flux due to this configuration through the surface S is :



(A) $\frac{5q}{\epsilon_0}$ (B) $\frac{4q}{\epsilon_0}$ (C) $\frac{3q}{\epsilon_0}$ (D) $\frac{q}{\epsilon_0}$

PEO347

6. Let there be a spherically symmetric charge distribution with charge density varying as

$\rho(r) = \rho_0 \left(\frac{5}{4} - \frac{r}{R} \right)$ upto $r = R$, and $\rho(r) = 0$ for $r > R$, where r is the distance from the origin. The

electric field at a distance r ($r < R$) from the origin is given by :

(A) $\frac{\rho_0 r}{3\epsilon_0} \left(\frac{5}{4} - \frac{r}{R} \right)$ (B) $\frac{4\pi\rho_0 r}{3\epsilon_0} \left(\frac{5}{3} - \frac{r}{R} \right)$ (C) $\frac{\rho_0 r}{4\epsilon_0} \left(\frac{5}{3} - \frac{r}{R} \right)$ (D) $\frac{4\rho_0 r}{3\epsilon_0} \left(\frac{5}{4} - \frac{r}{R} \right)$

PEO348

7. Let a total charge $2Q$ be distributed in a sphere of radius R , with the charge density given by $\rho(r) = kr$, where r is the distance from the centre. Two charges A and B , of $-Q$ each, are placed on diametrically opposite points, at equal distance, a , from the centre. If A and B do not experience any force, then:

(A) $a = \frac{3R}{2^{3/4}}$ (B) $a = R/\sqrt{3}$ (C) $a = 8^{-1/4}R$ (D) $a = 2^{-1/4}R$

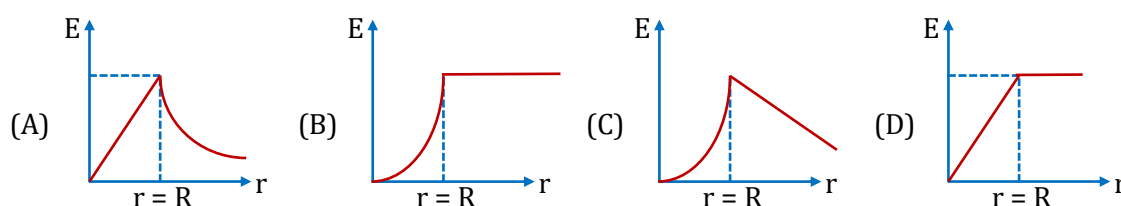
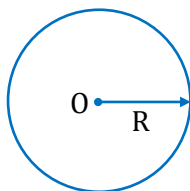
PEO349

8. In finding the electric field using Gauss Law the formula $|\vec{E}| = \frac{q_{enc}}{\epsilon_0 A}$ is applicable. In the formula ϵ_0 is permittivity of free space, A is the area of Gaussian surface and q_{enc} is charge enclosed by the Gaussian surface. The equation can be used in which of the following situation?

- (A) Only when the Gaussian surface is an equipotential surface.
 (B) Only when $|\vec{E}| = \text{constant}$ on the surface.
 (C) For any choice of Gaussian surface.
 (D) Only when the Gaussian surface is an equipotential surface and $|\vec{E}|$ is constant on the surface.

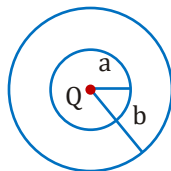
PEO350

9. Graphical variation of electric field due to a uniformly charged insulating solid sphere of radius R , with distance r from the centre O is represented by:



PEO351

10. The region between two concentric spheres of radii ' a ' and ' b ', respectively (see figure), has volume charge density $\rho = \frac{A}{r}$, where A is a constant and r is the distance from the centre. At the centre of the spheres is a point charge Q . The value of A such that the electric field in the region between the spheres will be constant, is:-



(A) $\frac{2Q}{\pi a^2}$

(B) $\frac{Q}{2\pi a^2}$

(C) $\frac{Q}{2\pi(b^2 - a^2)}$

(D) $\frac{2Q}{\pi(a^2 - b^2)}$

PE0352

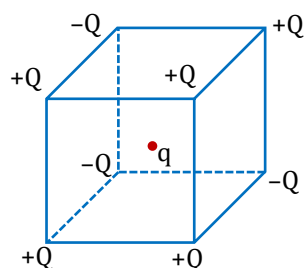
ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	D	12	288	B	B	C	C	D	A	B



Electrostatics : RACE-05

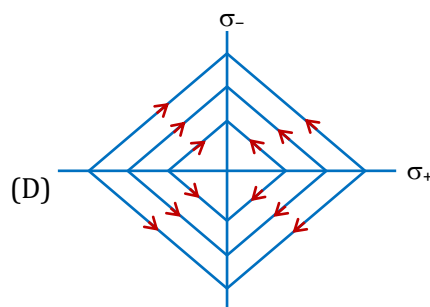
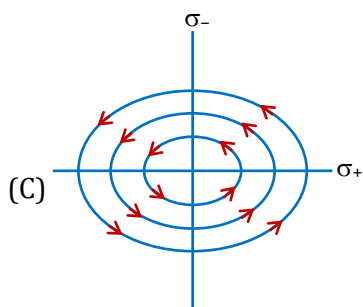
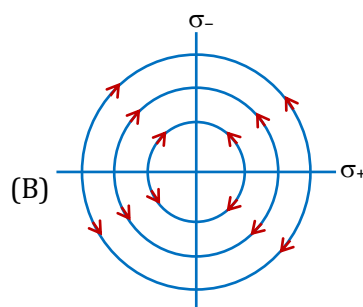
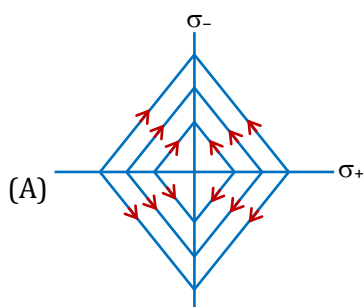
1. Nine point charges are placed on a cube as shown in the figure. The charge q is placed at the body centre whereas all other charges are at the vertices. The electrostatic potential energy of the system will be :-



- (A) Positive (B) Negative (C) Zero (D) Depends on sign and value of q

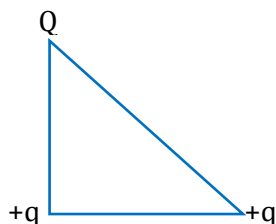
PE0353

2. Two charged thin infinite plane sheets of uniform surface charge density σ_+ and σ_- where $|\sigma_+| > |\sigma_-|$ intersect at right angle. Which of the following best represents the electric field lines for this system :



PE0354

3. The charges Q , $+q$ and $+q$ are placed at the vertices of a right-angle isosceles triangle as shown below. The net electrostatic energy of the configuration is zero, if the value of Q is:



- (A) $\frac{-\sqrt{2}q}{\sqrt{2}+1}$ (B) $-2q$ (C) $\frac{-q}{1+\sqrt{2}}$ (D) $+q$

PEO355

4. Four equal point charges Q each are placed in the xy plane at $(0, 2)$, $(4, 2)$, $(4, -2)$ and $(0, -2)$. The work required to put a fifth charge Q at the origin of the coordinate system will be :

- (A) $\frac{Q^2}{2\sqrt{2}\pi\epsilon_0}$ (B) $\frac{Q^2}{4\pi\epsilon_0}\left(1+\frac{1}{\sqrt{5}}\right)$ (C) $\frac{Q^2}{4\pi\epsilon_0}\left(1+\frac{1}{\sqrt{3}}\right)$ (D) $\frac{Q^2}{4\pi\epsilon_0}$

PEO356

5. The electric field in a region is given by $\vec{E} = (Ax + B)\hat{i}$, where E is in NC^{-1} and x is in metres. The values of constants are $A = 20$ SI unit and $B = 10$ SI unit. If the potential at $x = 1$ is V_1 and that at $x = -5$ is V_2 , then $V_1 - V_2$ is :-

- (A) -48 V (B) -520 V (C) 180 V (D) 320 V

PEO357

6. A particle of charge q and mass m is subjected to an electric field $E = E_0(1 - ax^2)$ in the x -direction, where a and E_0 are constants. Initially the particle was at rest at $x = 0$. Other than the initial position the kinetic energy of the particle becomes zero when the distance of the particle from the origin is:

- (A) $\sqrt{\frac{2}{a}}$ (B) $\sqrt{\frac{1}{a}}$ (C) a (D) $\sqrt{\frac{3}{a}}$

PEO358

7. If the electric potential at any point (x, y, z) m in space is given by $V = 3x^2$ volt. The electric field at the point $(1, 0, 3)$ m will be

- (A) 3 Vm^{-1} , directed along positive x -axis. (B) 3 Vm^{-1} , directed along negative x -axis.
(C) 6 Vm^{-1} , directed along positive x -axis. (D) 6 Vm^{-1} , directed along negative x -axis.

PEO359

8. An electric charge $10^{-6}\mu C$ is placed at origin $(0, 0)$ m of $X-Y$ co-ordinate system. Two points P and Q are situated at $(\sqrt{3}, \sqrt{3})$ m and $(\sqrt{6}, 0)$ m respectively. The potential difference between the points P and Q will be :

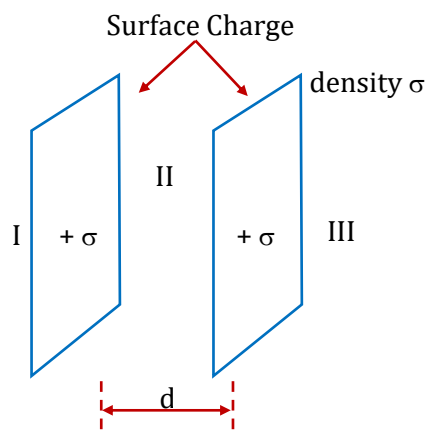
- (A) $\sqrt{3}V$ (B) $\sqrt{6}V$ (C) 0 V (D) 3 V

PEO360

9. The electric potential at the surface of an atomic nucleus ($z = 50$) of radius 9×10^{-13} cm is _____ $\times 10^6$ V.

PEO361

10. Let σ be the uniform surface charge density of two infinite thin plane sheets shown in figure. Then the electric fields in three different region E_I , E_{II} and E_{III} are:



- (A) $\vec{E}_I = \frac{2\sigma}{\epsilon_0} \hat{n}, \vec{E}_{II} = 0, \vec{E}_{III} = \frac{2\sigma}{\epsilon_0} \hat{n}$ (B) $\vec{E}_I = 0, \vec{E}_{II} = \frac{\sigma}{\epsilon_0} \hat{n}, \vec{E}_{III} = 0$
- (C) $\vec{E}_I = \frac{\sigma}{2\epsilon_0} \hat{n}, \vec{E}_{II} = 0, \vec{E}_{III} = \frac{\sigma}{2\epsilon_0} \hat{n}$ (D) $\vec{E}_I = -\frac{\sigma}{\epsilon_0} \hat{n}, \vec{E}_{II} = 0, \vec{E}_{III} = \frac{\sigma}{\epsilon_0} \hat{n}$

PEO362

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	D	A	A	B	C	D	D	C	8.00	D



Electrostatics : RACE-06

1. A uniformly charged ring of radius $3a$ and total charge q is placed in xy -plane centred at origin. A point charge q is moving towards the ring along the z -axis and has speed u at $z = 4a$. The minimum value of u such that it crosses the origin is :

(A) $\sqrt{\frac{2}{m} \left(\frac{1}{15} \frac{q^2}{4\pi\epsilon_0 a} \right)^{1/2}}$ (B) $\sqrt{\frac{2}{m} \left(\frac{2}{15} \frac{q^2}{4\pi\epsilon_0 a} \right)^{1/2}}$ (C) $\sqrt{\frac{2}{m} \left(\frac{4}{15} \frac{q^2}{4\pi\epsilon_0 a} \right)^{1/2}}$ (D) $\sqrt{\frac{2}{m} \left(\frac{1}{5} \frac{q^2}{4\pi\epsilon_0 a} \right)^{1/2}}$

PEO363

2. In free space, a particle A of charge $1 \mu\text{C}$ is held fixed at a point P . Another particle B of the same charge and mass $4 \mu\text{g}$ is kept at a distance of 1 mm from P . if B is released, then its velocity at a distance of 9 mm from P is:

[Take $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2 \text{ C}^{-2}$]

(A) $2.0 \times 10^3 \text{ m/s}$ (B) $3.0 \times 10^4 \text{ m/s}$ (C) $1.5 \times 10^2 \text{ m/s}$ (D) 1.0 m/s

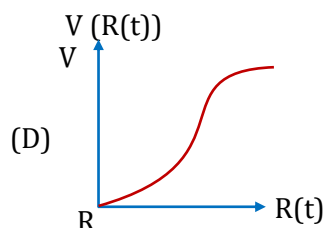
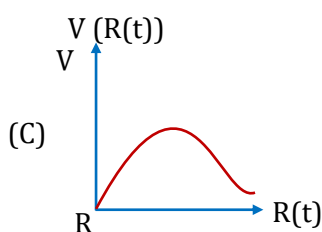
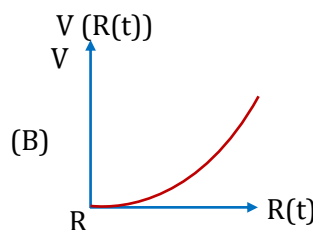
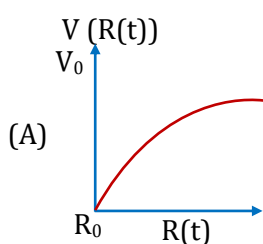
PEO364

3. The two thin coaxial rings, each of radius ' a ' and having charges $+Q$ and $-Q$ respectively are separated by a distance of ' s '. The potential difference between the centres of the two rings is :

(A) $\frac{Q}{2\pi\epsilon_0} \left[\frac{1}{a} + \frac{1}{\sqrt{s^2 + a^2}} \right]$ (B) $\frac{Q}{4\pi\epsilon_0} \left[\frac{1}{a} + \frac{1}{\sqrt{s^2 + a^2}} \right]$
 (C) $\frac{Q}{4\pi\epsilon_0} \left[\frac{1}{a} - \frac{1}{\sqrt{s^2 + a^2}} \right]$ (D) $\frac{Q}{2\pi\epsilon_0} \left[\frac{1}{a} - \frac{1}{\sqrt{s^2 + a^2}} \right]$

PEO365

4. There is a uniform spherically symmetric surface charge density at a distance R_0 from the origin. The charge distribution is initially at rest and starts expanding because of mutual repulsion. The figure that represents best the speed $V(R(t))$ of the distribution as a function of its instantaneous radius $R(t)$ is :

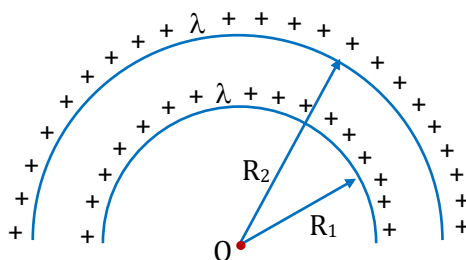


PEO366

5. An electric field $\vec{E} = (25\hat{i} + 30\hat{j}) \text{ NC}^{-1}$ exists in a region of space. If the potential at the origin is taken to be zero then the potential at $x = 2\text{m}, y = 2\text{m}$ is :-
 (A) -140 J (B) -120 J (C) -110 J (D) -130 J

PEO367

6. The electric potential at the centre of two concentric half rings of radii R_1 and R_2 , having same linear charge density λ is



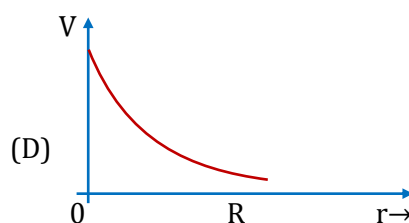
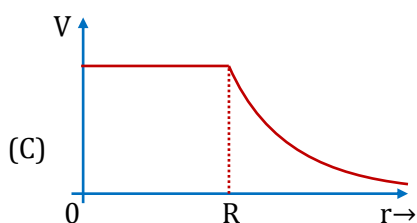
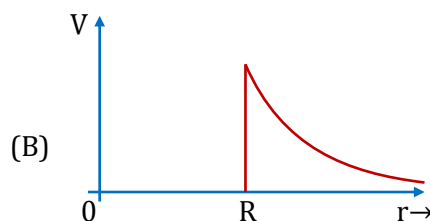
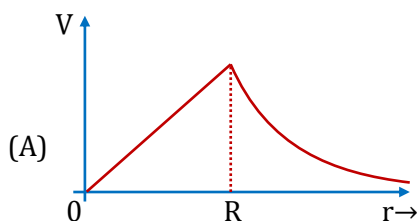
- (A) $\frac{2\lambda}{\epsilon_0}$ (B) $\frac{\lambda}{2\epsilon_0}$ (C) $\frac{\lambda}{4\epsilon_0}$ (D) $\frac{\lambda}{\epsilon_0}$

PEO368

7. A uniformly charged solid sphere of radius R has potential V_0 (measured with respect to ∞) on its surface. For this sphere the equipotential surfaces with potentials $\frac{3V_0}{2}, \frac{5V_0}{4}, \frac{3V_0}{4}$ and $\frac{V_0}{4}$ have radius R_1, R_2, R_3 and R_4 respectively. Then
 (A) $R_1 = 0$ and $R_2 < (R_4 - R_3)$ (B) $2R < R_4$
 (C) $R_1 = 0$ and $R_2 > (R_4 - R_3)$ (D) $R_1 \neq 0$ and $(R_2 - R_1) > (R_4 - R_3)$

PEO369

8. Which of the following correctly represents the variation of electric potential (V) of a charged spherical conductor of radius (R) with radial distance (r) from the centre ?



PEO370

9. 64 identical drops each charged upto potential of 10 mV are combined to form a bigger drop. The potential of the bigger drop will be _____ mV .

PEO371

10. Consider two charged metallic spheres S_1 and S_2 of radii R_1 and R_2 , respectively. The electric fields E_1 (on S_1) and E_2 (on S_2) on their surfaces are such that $E_1/E_2 = R_1/R_2$. Then the ratio V_1 (on S_1) / V_2 (on S_2) of the electrostatic potentials on each sphere is :

- (A) (R_2/R_1) (B) $\left(\frac{R_1}{R_2}\right)^3$ (C) R_1/R_2 (D) $(R_1/R_2)^2$

PE0372

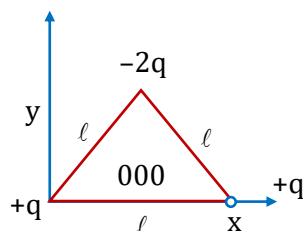
ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	A	D	A	C	B	A or B	C	160	D



Electrostatics : RACE-07

1. Determine the electric dipole moment of the system of three charges, placed on the vertices of an equilateral triangle, as shown in the figure:



- (A) $(q\ell)\frac{\hat{i}+\hat{j}}{\sqrt{2}}$ (B) $\sqrt{3}q\ell\frac{\hat{j}-\hat{i}}{\sqrt{2}}$ (C) $-\sqrt{3}q\ell\hat{j}$ (D) $2q\ell\hat{j}$

PEO373

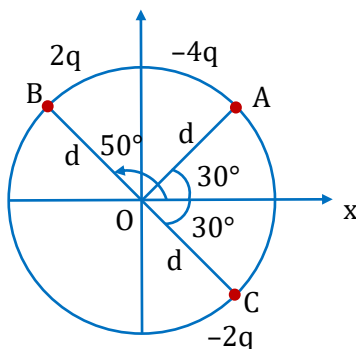
2. A point dipole $\vec{p} = -p_0\hat{x}$ is kept at the origin. The potential and electric field due to this dipole on the y-axis at a distance d are, respectively:

(Take $V = 0$ at infinity) :

- (A) $\frac{|\vec{p}|}{4\pi\epsilon_0 d^2}, \frac{-\vec{p}}{4\pi\epsilon_0 d^3}$ (B) $0, \frac{\vec{p}}{4\pi\epsilon_0 d^3}$ (C) $\frac{|\vec{p}|}{4\pi\epsilon_0 d^2}, \frac{\vec{p}}{4\pi\epsilon_0 d^3}$ (D) $0, \frac{-\vec{p}}{4\pi\epsilon_0 d^3}$

PEO374

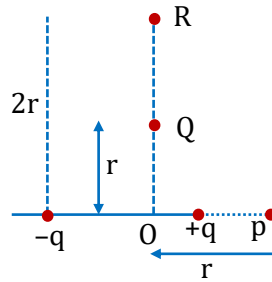
3. Three charged particle A, B and C with charges $-4q, 2q$ and $-2q$ are present on the circumference of a circle of radius d . the charged particles A, C and centre O of the circle formed an equilateral triangle as shown in figure. Electric field at O along x-direction is :



- (A) $\frac{2\sqrt{3}q}{\pi\epsilon_0 d^2}$ (B) $\frac{\sqrt{3}q}{4\pi\epsilon_0 d^2}$ (C) $\frac{3\sqrt{3}q}{4\pi\epsilon_0 d^2}$ (D) $\frac{\sqrt{3}q}{\pi\epsilon_0 d^2}$

PEO375

4. The electric field at point p due to an electric dipole is E . The electric field at point R on equatorial line will be $\frac{E}{x}$. The value of x :



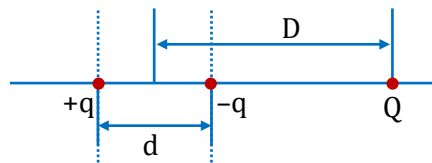
PEO376

5. Four-point charges $-q, +q, +q$ and $-q$ are placed on y -axis at $y = -2d, y = -d, y = +d$ and $y = +2d$, respectively. The magnitude of the electric field E at a point on the x -axis at $x = D$, with $D \gg d$, will behave as :-

(A) $E \propto \frac{1}{D}$ (B) $E \propto \frac{1}{D^3}$ (C) $E \propto \frac{1}{D^2}$ (D) $E \propto \frac{1}{D^4}$

PEO377

6. A system of three charges are placed as shown in the figure :

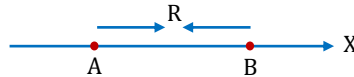


If $D \gg d$, the potential energy of the system is best given by :

(A) $\frac{1}{4\pi\epsilon_0} \left[-\frac{q^2}{d} - \frac{qQd}{2D^2} \right]$ (B) $\frac{1}{4\pi\epsilon_0} \left[+\frac{q^2}{d} + \frac{qQd}{D^2} \right]$
 (C) $\frac{1}{4\pi\epsilon_0} \left[-\frac{q^2}{d} + \frac{2qQd}{D^2} \right]$ (D) $\frac{1}{4\pi\epsilon_0} \left[-\frac{q^2}{d} - \frac{qQd}{D^2} \right]$

PEO378

7. Two electric dipoles, A, B with respective dipole moments $\vec{d}_A = -4qa\hat{i}$ and $\vec{d}_B = -2qa\hat{i}$ placed on the x -axis with a separation R , as shown in the figure:



The distance from A at which both of them produce the same potential is :

(A) $\frac{\sqrt{2}R}{\sqrt{2}+1}$ (B) $\frac{R}{\sqrt{2}+1}$ (C) $\frac{\sqrt{2}R}{\sqrt{2}-1}$ (D) $\frac{R}{\sqrt{2}-1}$

PEO379

8. An electric dipole of moment $\vec{p} = (-\hat{i} - 3\hat{j} + 2\hat{k}) \times 10^{-29} \text{ C.m}$ is at the origin $(0, 0, 0)$. The electric field due to this dipole at $\vec{r} = +\hat{i} + 3\hat{j} + 5\hat{k}$ (note that $\vec{r} \cdot \vec{p} = 0$) is parallel to:

(A) $(-\hat{i} + 3\hat{j} - 2\hat{k})$ (B) $(+\hat{i} - 3\hat{j} - 2\hat{k})$ (C) $(+\hat{i} + 3\hat{j} - 2\hat{k})$ (D) $(-\hat{i} - 3\hat{j} + 2\hat{k})$

PEO380

9. An electric dipole is formed by two equal and opposite charges q with separation d . The charges have same mass m . It is kept in a uniform electric field E . If it is slightly rotated from its equilibrium orientation, then its angular frequency ω is :-

(A) $\sqrt{\frac{qE}{2md}}$ (B) $2\sqrt{\frac{qE}{md}}$ (C) $\sqrt{\frac{2qE}{md}}$ (D) $\sqrt{\frac{qE}{md}}$

PEO381

10. An electric dipole of dipole moment is $6.0 \times 10^{-6} \text{ Cm}$ placed in a uniform electric field of $1.5 \times 10^3 \text{ NC}^{-1}$ in such a way that dipole moment is along electric field. The work done in rotating dipole by 180° in this field will be _____ mJ .

PEO382

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	C	D	D	16	D	D	C	C	C	18



Electrostatics : RACE-08

1. Given below are two statements:

Statement I : Electric potential is constant within and at the surface of each conductor.

Statement II : Electric field just outside a charged conductor is perpendicular to the surface of the conductor at every point.

In the light of the above statements, choose the most appropriate answer from the options give below.

- (A) Both statement I and statement II are correct.
 (B) Both statement I and statement II are incorrect.
 (C) Statement I is correct but statement II is incorrect.
 (D) Statement I is incorrect but and statement II is correct.

PEO383

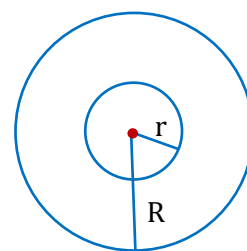
2. A charge Q is distributed over three concentric spherical shells of radii a, b, c ($a < b < c$) such that their surface charge densities are equal to one another. The total potential at a point at distance r from their common centre, where $r < a$, would be:

- (A) $\frac{Q}{4\pi\epsilon_0(a+b+c)}$ (B) $\frac{Q(a+b+c)}{4\pi\epsilon_0(a^2+b^2+c^2)}$
 (C) $\frac{Q}{12\pi\epsilon_0} \frac{ab+bc+ca}{abc}$ (D) $\frac{Q}{4\pi\epsilon_0} \frac{(a^2+b^2+c^2)}{(a^3+b^3+c^3)}$

PEO384

3. A charge Q is distributed over two concentric conducting thin spherical shells radii r and R ($R > r$). If the surface charge densities on the two shells are equal, the electric potential at the common centre is :

- (A) $\frac{1}{4\pi\epsilon_0} \frac{(R+2r)Q}{2(R^2+r^2)}$ (B) $\frac{1}{4\pi\epsilon_0} \frac{(R+r)}{2(R^2+r^2)} Q$
 (C) $\frac{1}{4\pi\epsilon_0} \frac{(R+r)}{(R^2+r^2)} Q$ (D) $\frac{1}{4\pi\epsilon_0} \frac{(2R+r)}{(R^2+r^2)} Q$



PEO385

4. Two isolated conducting spheres S_1 and S_2 of radius $\frac{2}{3}R$ and $\frac{1}{3}R$ have $12 \mu C$ and $-3 \mu C$ charges, respectively, and are at a large distance from each other. They are now connected by a conducting wire. A long time after this is done the charges on S_1 and S_2 are respectively :

- (A) $6 \mu C$ and $3 \mu C$ (B) $+4.5 \mu C$ and $-4.5 \mu C$
 (C) $3 \mu C$ and $6 \mu C$ (D) $4.5 \mu C$ on both

PEO386

5. Consider the force F on a charge ' q ' due to a uniformly charged spherical shell of radius R carrying charge Q distributed uniformly over it. Which one of the following statements is true for F , if ' q ' is placed at distance r from the centre of the shell ?

(A) $F = \frac{1}{4\pi\epsilon_0} \frac{Qq}{r^2}$ for $r > R$ (B) $\frac{1}{4\pi\epsilon_0} \frac{qQ}{R^2} > F > 0$ for $r < R$
 (C) $F = \frac{1}{4\pi\epsilon_0} \frac{Qq}{r^2}$ for all r (D) $F = \frac{1}{4\pi\epsilon_0} \frac{Qq}{R^2}$ for $r < R$

PEO387

6. σ is the uniform surface charge density of a thin spherical shell of radius R . The electric field at any point on the surface of the spherical shell is :

(A) $\sigma/\epsilon_0 R$ (B) $\sigma/2\epsilon_0$ (C) σ/ϵ_0 (D) $\sigma/4\epsilon_0$

PEO388

7. The vehicles carrying inflammable fluids usually have metallic chains touching the ground :

- (A) To conduct excess charge due to air friction to ground and prevent sparking.
 (B) To alert other vehicles.
 (C) To protect tyres from catching dirt from ground.
 (D) It is a custom.

PEO389

8. Two charged conducting spheres of radii a and b are connected to each other by a conducting wire. The ratio of charges of the two spheres respectively is:

(A) $\sqrt{ab}$ (B) ab (C) $\frac{a}{b}$ (D) $\frac{b}{a}$

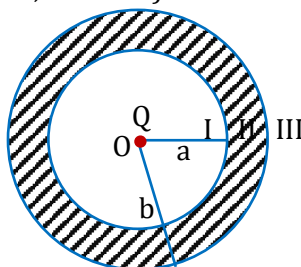
PEO390

9. Two isolated metallic solid spheres of radii R and $2R$ are charged such that both have same charge density σ . The spheres are then connected by a thin conducting wire. If the new charge density of the bigger sphere is σ' . The ratio $\frac{\sigma'}{\sigma}$ is :

(A) $\frac{9}{4}$ (B) $\frac{4}{3}$ (C) $\frac{5}{3}$ (D) $\frac{5}{6}$

PEO391

10. As shown in the figure, a point charge Q is placed at the centre of conducting spherical shell of inner radius a and outer radius b . The electric field due to charge Q in three different regions I, II and III is given by : (I : $r < a$, II : $a < r < b$, III : $r > b$)



- (A) $E_I = 0, E_{II} = 0, E_{III} \neq 0$ (B) $E_I \neq 0, E_{II} = 0, E_{III} \neq 0$
 (C) $E_I \neq 0, E_{II} = 0, E_{III} = 0$ (D) $E_I = 0, E_{II} = 0, E_{III} = 0$

PEO392

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	B	C	A	A	C	A	C	D	B